Snake Game Learning Journal - Session 2

1. Snake Representation

Instead of storing one position, the snake is stored as a list of body segments. Example: snake = [[340,200],[320,200],[300,200]]. The first element is always the head.

2. Accessing the Head

snake[0] returns the head. head_x = snake[0][0] and head_y = snake[0][1].

3. Creating a New Head

new_head = [head_x + dx, head_y + dy]. This calculates where the snake will move next.

4. insert()

snake.insert(0, new_head) inserts the new head at the front of the list.

5. pop()

snake.pop() removes the last segment (the tail).

6. Snake Movement Algorithm

Get the head, calculate a new head, insert it at the front, and remove the tail.

7. Snake Growth

If the snake eats food, do NOT call snake.pop(). The snake grows by one segment.

8. if / else Logic

If the snake eats food, generate new food. Otherwise remove the tail using snake.pop().

9. Correct Game Loop

1. Handle input. 2. Update game state. 3. Draw everything. 4. Update the display. 5. Control frame rate.

10. Git & GitHub

Learned git init, git add ., git commit -m "Initial Snake game", and git push. Successfully created the first GitHub repository and pushed the Snake project.

Key Lesson

Programming is about understanding algorithms, not memorizing syntax. The movement algorithm is: Get head -> Calculate new head -> Insert new head -> Remove tail.